

3.1 Functions of Two or More Variables Worksheet

1. For each of the following functions state the domain & range in set notation and create a picture of the domain in the xy -plane.

a) $f(x, y) = \sqrt{x} - \ln(y)$.

b) $f(x, y) = \sqrt{x^2 - y}$

c) $f(x, y) = \tan(x) + y$

d) $f(x, y) = \frac{\sqrt{y - x^2}}{1 - x^2}$

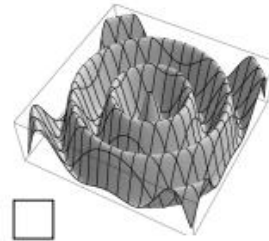
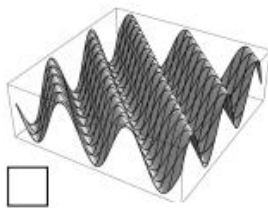
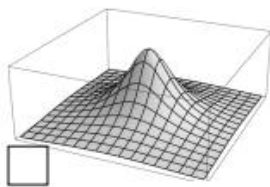
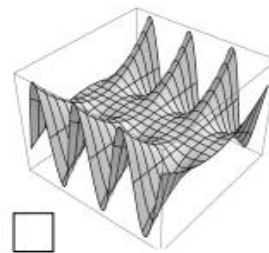
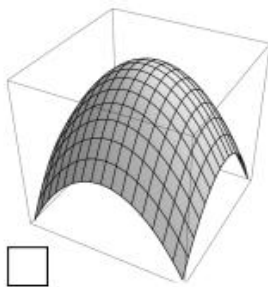
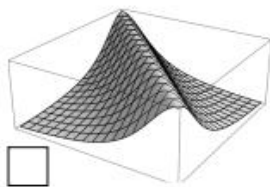
e) $f(x, y) = \arcsin(x^2 + y^2 - 2)$

f) $f(x, y) = \ln(9 - x^2 - 9y^2)$

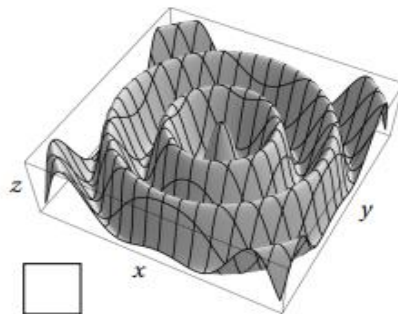
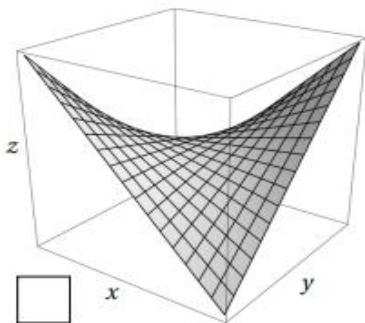
2. For each function, label its graph from among the options below by writing the corresponding letter in the box next to the graph.

(A) $z = -y^2 \sin(x)$

(B) $z = \frac{1}{1+x^2+y^2}$



3. For each graph below, find the corresponding function from the options at the right and write the corresponding letter in the box next to the graph.



(A) $\sin(x+y)$

(B) $x^2 - y^2$

(C) $\cos(\sqrt{x^2+y^2})$

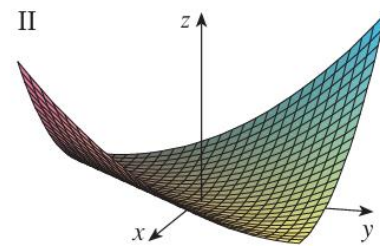
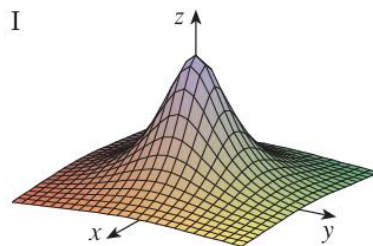
(D) $\cos(x)\cos(y)$

(E) $(x-y)^2$

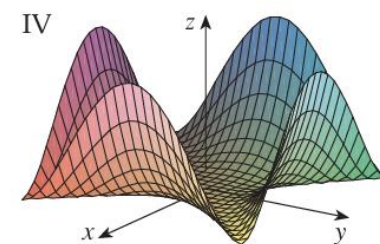
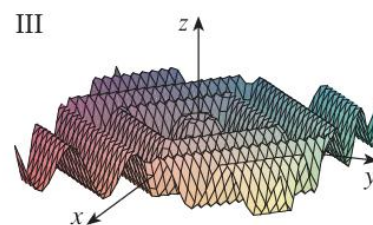
(F) xy

4. Match the function with its graph.

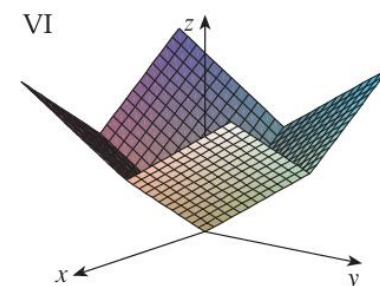
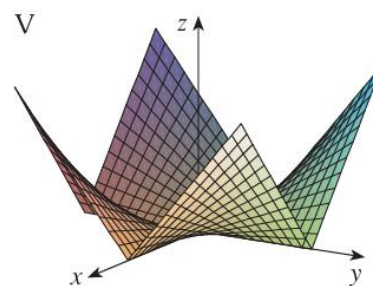
a) $f(x, y) = |x| + |y|$



b) $f(x, y) = |xy|$



c) $f(x, y) = \frac{1}{1 + x^2 + y^2}$



d) $f(x, y) = (x^2 - y^2)^2$

e) $f(x, y) = (x - y)^2$

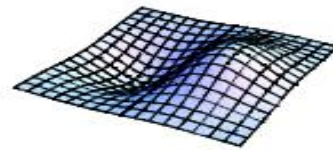
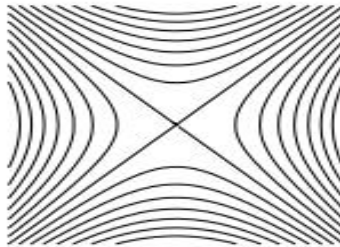
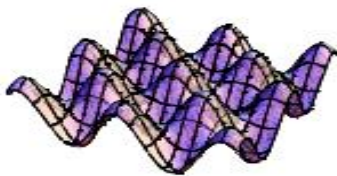
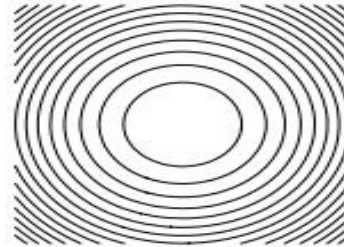
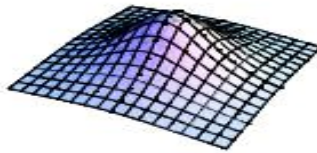
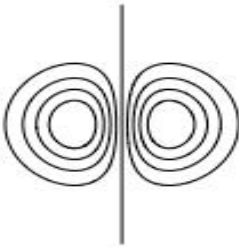
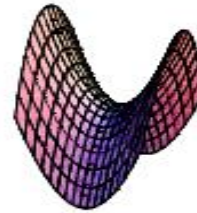
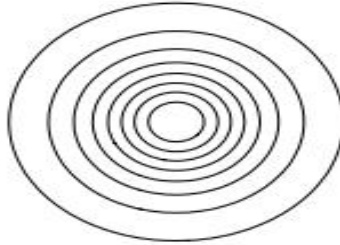
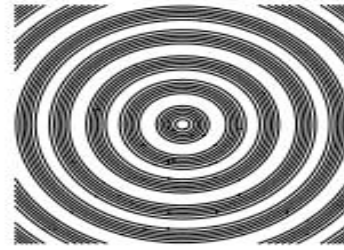
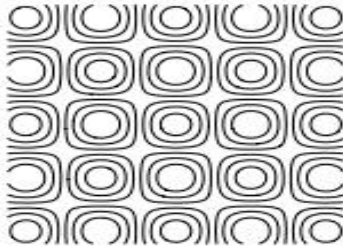
f) $f(x, y) = \sin(|x| + |y|)$

5. Match the following functions with their graphs and level set diagrams.

a) $f(x, y) = \sin(x)\cos(y)$

b) $f(x, y) = x^2 + y^2$

c) $f(x, y) = x^2 - y^2$



6. Match the function with its graph and with its contour map.

a) $z = \sin(xy)$

b) $z = e^x \cos(y)$

c) $z = \sin(x - y)$

d) $z = \sin(x) - \sin(y)$

e) $z = (1 - x^2)(1 - y^2)$

f) $z = \frac{x - y}{1 + x^2 + y^2}$

